

AN IMPROVED LITTLEWOOD CONSTANT AND STRUCTURAL INVERSE THEOREMS

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ABSTRACT. For a finite set $A \subset \mathbb{Z}$, write $\mathcal{L}(A) = \int_0^1 |\sum_{a \in A} e^{2\pi i a t}| dt$. We prove the uniform lower bound $\mathcal{L}(A) \geq (c_* - o(1)) \log |A|$, where $c_* > 0.2459209$ is explicit. The argument refines the McGehee–Pigno–Smith construction analysed by Bloom and Green: outer multipliers use an exact pointwise modulus budget, and a whole-product estimate couples their error to the loss of squared modulus. All even moments are controlled by interval moments. We also prove a structural inverse theorem for an explicit class of integer sets generated from dense separated blocks and long arithmetic block inflations by cyclic Sidon-fibre decompositions of arbitrary depth. If $\mathcal{L}(A) \leq K \log |A|$, then, for every $\varepsilon > 0$, all but $\varepsilon|A|$ points are covered by boundedly many disjoint, linear-sized subsets of bounded doubling. The constants are independent of the decomposition depth, branching, moduli, and diameter. The proof combines a constant-one Sidon coefficient inequality with hereditary energy estimates obtained from bounded majority amplification and integer boundaries. The conclusion survives high-order Freiman modelling. The decomposition hypothesis is substantive; no unrestricted linear-sized inverse theorem or sharp $4/\pi^2$ lower bound is asserted.

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1. INTRODUCTION

Let $e(t) = \exp(2\pi it)$ and, for a finite set of integers A , put

$$F_A(t) = \sum_{a \in A} e(at), \quad \mathcal{L}(A) = \int_{\mathbb{T}} |F_A(t)| dt, \quad \mathbb{T} = \mathbb{R}/\mathbb{Z}. \quad (1.1)$$

The measure on $\mathbb{T}$ is the probability Haar measure, and all logarithms in this paper are natural. Littlewood's conjecture, proved independently by Konyagin [11] and McGehee, Pigno, and Smith [14], states that $\mathcal{L}(A) \gg \log |A|$. The order of magnitude is attained by arithmetic progressions. Indeed, if P is a progression of length N , then

$$\mathcal{L}(P) = \frac{4}{\pi^2} \log N + O(1). \quad (1.2)$$

Translation and nonzero integer dilation preserve $\mathcal{L}$, so this expression does not depend on the location or common difference of P .

There are two natural questions beyond the logarithmic lower bound. The first is quantitative: can its leading constant be increased towards $4/\pi^2$? The second is structural: what can a set look like when its Fourier L^1 norm is only a constant multiple of $\log N$, rather than a quantity comparable to the elementary upper bound $\sqrt{N}$? These questions are related through duality, but a quantitative improvement to a bounded test function does not by itself provide a structural inverse theorem. This paper treats both questions, and keeps their conclusions and hypotheses separate.

1.1. Previous work and the connection with Bloom–Green. The stronger assertion that an arithmetic progression minimizes $\mathcal{L}(A)$ among all N -element sets, as well as its leading-constant version, is usually called the strong Littlewood conjecture; see Hardy–Littlewood [10] and Zygmund [21]. The MPS proof gives a weighted inequality for arbitrary complex coefficients, not merely the indicator-set case. Stegeman [18] and Yabuta [20] refined the numerical constant in that argument, with Yabuta obtaining a coefficient approximately 0.12959. Fournier [6] discusses the bounded-test-function methods underlying the classical proofs.

Bloom and Green [2] revisit the MPS construction with two purposes. They obtain the coefficient $0.170934\dots$, and, by examining the failure of a putative larger lower bound, prove the following inverse statement. If $|A| = N$ and $\mathcal{L}(A) \leq K \log N$, then for $0 < \delta \leq 1/2$ there is an initial segment $A' \subseteq A$ with

$$|A'| \gg N^{1-\delta}, \quad E(A') \gg (\delta/K)^2 |A'|^3, \quad (1.3)$$

where

$$E(B) = \#\{(b_1, b_2, b_3, b_4) \in B^4 : b_1 + b_2 = b_3 + b_4\} = \int_{\mathbb{T}} |F_B|^4$$

is the additive energy. Their analysis exposes an energy square root in the error of the ordered MPS recursion. Standard additive-combinatorial results then turn (1.3) into small-doubling subsets and arithmetic-progression consequences. Their result is unrestricted: it applies to every finite integer set.

Earlier structural work addressed rather different descriptions of complexity. Petridis [15] and Hanson [9] show that sufficiently genuine multidimensional structure increases the Fourier norm. Pichorides [16], and independently Bedert [1], relate the norm to the size of dissociated subsets. In another direction, the finite-group theorem of Green and Sanders [8] classifies integer-valued functions of bounded Fourier algebra norm by bounded combinations of coset indicators. The latter has a constant spectral budget, unlike the logarithmic budget of the integer Littlewood problem. Bourgain [3] and Bedert [1] also connect these questions to large sum-free subsets.

Our starting point is the same dual principle as in Bloom–Green: a bounded test function tries to accumulate contributions, and the obstruction to accumulation contains information. For the numerical problem we retain the entire product of the damping multipliers. For the inverse problem we arrange that an obstruction is inherited by every remaining family of blocks, while all pairings continue to use the *original* target. This distinction allows repeated structural extraction without assuming that deleting frequencies decreases an L^1 norm.

1.2. The quantitative theorem. For $p > 1$ define

$$I_p = \int_{\mathbb{R}} \left| \frac{\sin \pi x}{\pi x} \right|^p dx, \quad c_{2r} = I_{2r}, \quad c_{2r+1} = \sqrt{I_{2r} I_{2r+2}} \quad (r \geq 1). \quad (1.4)$$

Put $a = 9/10$ and

$$\mathcal{E}_a = \frac{55a^2}{108} + \frac{1}{9} \left\{ \frac{2}{3} \left(\frac{5a^4}{6} - \frac{2a^3}{3} \right) + 2 \sum_{k=5}^{\infty} \frac{c_k a^k}{k} \right\}, \quad (1.5)$$

$$c_* = \frac{9}{10 \log 9} \left(1 - \sqrt{\mathcal{E}_{9/10}/3} \right). \quad (1.6)$$

The series is absolutely convergent, since $0 < c_k \leq 1$.

Theorem 1.1 (An explicit Littlewood coefficient). *For every finite set $A \subset \mathbb{Z}$ of cardinality N ,*

$$\mathcal{L}(A) \geq (c_* - o(1)) \log N, \quad c_* > 0.2459209. \quad (1.7)$$

The $o(1)$ is uniform over all N -element integer sets.

The parameter choices in (1.5) are explicit rather than numerically optimized. The proof has three ingredients. First, an outer multiplier of modulus $1 - a|F_B|/|B|$ uses exactly the pointwise room left by the source term $aF_B/|B|$. Second, the error of a product is estimated before its factors are separated; its second-order expansion retains the loss of squared modulus. Third, the same scalar variable controls both terms, and interval moment bounds handle the only tail level at which a nonconstant majorant is needed. We provide a self-contained proof of the moment comparison using symmetric chains, and an exact-rational certificate for the displayed decimal.

1.3. The structural theorem and its precise class. The conclusion of our inverse theorem is an almost-covering by large small-doubling sets. Its hypotheses specify how the original integer set is assembled; they are not deduced from small $\mathcal{L}$. We give the complete definition here so that the scope of the main statement is visible from the introduction.

Definition 1.2 (Additive Sidon sets). A subset R of an abelian group is *Sidon* if $r_1 + r_2 = r_3 + r_4$ with $r_i \in R$ implies equality of the two unordered pairs, with repetitions allowed. This is the additive B_2 condition, not the harmonic-analytic definition of a Sidon spectrum.

Fix $0 < \beta \leq 1$ and $\Delta \geq 1$. We use two terminal geometries, together with singletons. Their affine copies are also allowed.

Definition 1.3 (Terminal sets). A nonempty finite integer set is a terminal set if it is a singleton, or an affine copy under $x \mapsto u + vx$, $v \in \mathbb{Z} \setminus \{0\}$, of one of the following sets.

(i) A *dense separated block set* C , of size n , has an exact representation

$$C = \bigcup_{i=1}^R (b_i + C_i), \quad C_i \subseteq [-D, D] \cap \mathbb{Z}, \quad (1.8)$$

where D, M are positive integers and

$$|C_i| \geq M \geq n^\beta, \quad D \leq \Delta M, \quad |b_i - b_j| > 4D \quad (i \neq j). \quad (1.9)$$

The centres need not lie in a common progression and the C_i need not be intervals.

(ii) A long arithmetic block inflation is

$$A_M(\mathcal{C}) = \{b + q(Mu + r) : (b, u) \in \mathcal{C}, 0 \leq r < M\}, \quad (1.10)$$

where $q, M \geq 1$, $\emptyset \neq \mathcal{C} \subset (\mathbb{Z}/q\mathbb{Z}) \times \mathbb{Z}$ is finite, b is represented in $\{0, \dots, q-1\}$, and $M \geq V := |\mathcal{C}|$. Its cardinality is MV .

Definition 1.4 (The assembly class). Let $\mathfrak{G}_{\beta, \Delta}$ consist of the nonempty finite integer sets represented by finite rooted trees with terminal sets as leaves. At every internal vertex, after an affine normalization, its set is the exact union

$$A_v = \bigcup_{r \in R_v} (r + q_v A_{v,r}), \quad q_v \geq 2, \quad (1.11)$$

where $R_v \subset \mathbb{Z}/q_v\mathbb{Z}$ is the *entire* nonempty residue support, is Sidon, and the children are the nonempty sets $A_{v,r}$. The images of the children partition the parent. One-child vertices are allowed. There is no bound on depth, branching numbers, moduli, integer diameter, or the number of leaves.

Theorem 1.5 (An almost-covering inverse theorem). *Fix $K \geq 1$, $0 < \varepsilon < 1/2$, $0 < \beta \leq 1$, and $\Delta \geq 1$. There are $N_0, c > 0$, and $C < \infty$, depending only on these parameters, with the following property. If*

$$A \in \mathfrak{G}_{\beta, \Delta}, \quad |A| = N \geq N_0, \quad \mathcal{L}(A) \leq K \log N,$$

then there are pairwise disjoint nonempty subsets $B_1, \dots, B_J \subseteq A$ such that

$$\left| A \setminus \bigcup_{j=1}^J B_j \right| \leq \varepsilon N, \quad J \leq C, \quad |B_j| \geq cN, \quad |B_j + B_j| \leq C|B_j|. \quad (1.12)$$

The constants do not depend on any of the tree parameters excluded in Definition 1.4. The same conclusion holds if an order- N Freiman-isomorphic image of A belongs to $\mathfrak{G}_{\beta, \Delta}$.

Here a Freiman isomorphism of order h is a bijection preserving in both directions all equalities between sums of at most h terms on each side, with equal numbers of terms and with repetitions. The last clause is justified by a uniform estimate for the Fourier norms of *all* corresponding subsets; no order-two invariance of the norm is assumed.

Corollary 1.6 (Macroscopic hereditary energy). *Fix K, β, Δ and $0 < \delta \leq 1$. Under the geometric or modelling hypothesis of Theorem 1.5, if N is sufficiently large and $\mathcal{L}(A) \leq K \log N$, every $Y \subseteq A$ with $|Y| \geq \delta N$ satisfies*

$$E(Y) \geq \rho_{K, \beta, \Delta, \delta} |Y|^3$$

for a positive constant independent of A, N, Y .

Theorem 1.5 has a stronger conclusion than (1.3) on the specified class, but has a stronger hypothesis as well. It is not an unrestricted improvement of the Bloom–Green inverse theorem. In particular, we do not prove that a small-norm integer set possesses the tree in Definition 1.4, even after high-order modelling. The pieces in (1.12) are small-doubling sets, not necessarily single progressions, and their number can depend on ε . Thus the result also stops short of the signed-progression conjecture of Petridis and Bloom–Green.

1.4. How the two arguments fit together. In both parts of the paper, the main issue is to keep the information that a norm estimate would otherwise discard. In the quantitative part, separately maximizing the outer-function error and the surviving modulus loses their dependence on the same variable. The whole-product estimate restores that dependence. In

the structural part, replacing all failed amplifications by a single energy estimate loses the possibility of iteration. We instead prove an ambient-target hereditary inequality

$$E(S) \geq \eta \frac{|S|^4}{|\Lambda|} \quad (\emptyset \neq S \subseteq \Lambda) \quad (1.13)$$

for the integer labels of separated blocks. Bounded majority polynomials force a positive density of mixed correlations, and finite Fourier bandwidth turns those correlations into relations with a bounded carry. The original target is never replaced by the remaining block union.

Long arithmetic blocks provide a different route to the same invariant. Taking a discrete boundary removes the long direction, and the coefficient of its logarithmic contribution is a constant-scale spectral norm. Integer-valuedness then bounds the number of common boundary heights. Finite-group structure can be lifted back to integers while retaining the possible carries. Both local arguments feed into a hereditary energy-to-covering principle based on BSG.

A final obstruction to iteration is the multiplication of losses along an arbitrarily deep tree. We avoid it through the coefficient inequality

$$\left\| \sum_{r \in R} a_r \chi_r \right\|_1 \geq \left(\sum_{r \in R} |a_r|^{20} \right)^{1/20} \quad \text{for additive Sidon } R.$$

The exponent is not optimized; the constant one is essential. It gives an exact budget for all leaves at once, even for very unbalanced trees. Only boundedly many macroscopic leaves survive the logarithmic norm bound, and the two local covering theorems complete the proof.

Section 2 records the external inputs. Sections 3–4 prove Theorem 1.1. Sections 5–8 prove Theorem 1.5; Section 9 treats modelling and small exceptional perturbations. Examples and the remaining general problem are discussed in Section 10. The appendices give an exact numerical certificate and a one-sided obstruction, together with a two-sided interval model for the sharp constant.

2. NOTATION AND STANDARD INPUTS

For a function f on $\mathbb{T}$, its Fourier coefficient is $\widehat{f}(n) = \int_{\mathbb{T}} f(t) e(-nt) dt$. For finitely supported functions on an abelian group, convolution has its usual counting-measure convention. The transform F_X of an indicator is the sum of the associated characters on the compact dual. All dual Haar measures, including those on finite groups, have total mass one. Pairings are linear in the first entry: $\langle f, g \rangle = \int f \bar{g}$. The normalized energy is $\omega(X) = E(X)/|X|^3$. We use c, C for positive absolute constants that may change, with subscripts indicating additional dependence.

The identity $\int |F_X|^2 = |X|$ and Fourier inversion imply

$$1 \leq \mathcal{L}(X) \leq \sqrt{|X|} \quad (X \neq \emptyset). \quad (2.1)$$

Also, by Cauchy–Schwarz applied to the representation function of $X + X$,

$$E(X) \geq \frac{|X|^4}{|X + X|}. \quad (2.2)$$

These statements apply to finite subsets of any discrete abelian group.

Apart from standard Fourier and Hardy-space facts, we use the following two established results as external inputs. The further analytic and combinatorial implications needed for our main statements are proved below.

Theorem 2.1 (McGehee–Pigno–Smith [14]). *There is an absolute $c_{\text{MPS}} > 0$ such that, whenever $n_1 < \dots < n_m$ are integers and $u_1, \dots, u_m \in \mathbb{C}$,*

$$\int_{\mathbb{T}} \left| \sum_{j=1}^m u_j e(n_j t) \right| dt \geq c_{\text{MPS}} \sum_{j=1}^m \frac{|u_j|}{j}. \quad (2.3)$$

In particular, after decreasing c_{MPS} if necessary, $\mathcal{L}(X) \geq c_{\text{MPS}} \log(|X| + 1)$ for every nonempty finite integer set.

The complex-coefficient formulation (2.3) is also stated explicitly in Hanson [9, Theorem 1.1]. We will need it with some coefficients equal to zero, which is harmless: removing them only decreases their indices on the right.

Theorem 2.2 (Polynomial Balog–Szemerédi–Gowers). *There are absolute $c, C > 0$ such that, for a finite nonempty subset Y of an abelian group and $0 < \gamma \leq 1$, the inequality $E(Y) \geq \gamma|Y|^3$ yields a subset $Z \subseteq Y$ with*

$$|Z| \geq c\gamma^C|Y|, \quad |Z + Z| \leq C\gamma^{-C}|Z|.$$

We use this standard qualitative-polynomial form; see Tao–Vu [19, Theorem 2.31] and the quantitative refinements of Reiher–Schoen [17]. The latter are formulated with a small difference set. The standard sum–difference inequalities convert that conclusion to the displayed form, with a change in the absolute powers. No optimization of BSG exponents is used here.

For $q \geq 2$ and $r \in \{0, \dots, q-1\}$ set $A_r = \{m : r + qm \in A\}$. Character orthogonality gives

$$\frac{1}{q} \sum_{j=0}^{q-1} e(-rj/q) F_A(t + j/q) = e(rt) F_{A_r}(qt). \quad (2.4)$$

Consequently $\mathcal{L}(A_r) \leq \mathcal{L}(A)$. Moreover

$$\mathcal{L}(u + vA) = \mathcal{L}(A) \quad (u, v \in \mathbb{Z}, v \neq 0). \quad (2.5)$$

Both identities follow from Haar invariance, including for negative v . The contraction in (2.4) does not control $|A_r|$, and will not be used to discard a small fibre without a separate mass estimate.

3. INTERVAL BOUNDS FOR ALL EVEN MOMENTS

The numerical argument needs precise moment constants. Although these bounds are related to classical Fourier rearrangement inequalities [7, 10], for indicator sets they follow from an elementary antichain argument. We give the argument to make the precise extremal input, including its constant, independent of a rearrangement convention.

Lemma 3.1 (Symmetric chains). *The product $\{0, \dots, n-1\}^d$, partially ordered coordinatewise, has a partition into saturated chains symmetric about rank $d(n-1)/2$. In particular every antichain has size at most the largest rank level.*

Proof. A single chain has the required property. To pass from a symmetric chain decomposition of a ranked poset to one of its product with a chain, it suffices to decompose a rectangle $\{0, \dots, p\} \times \{0, \dots, q\}$. Assume $p \geq q$, interchanging the coordinates otherwise. For $0 \leq j \leq q$, take the chain

$$\{(i, j) : 0 \leq i \leq p-j\} \cup \{(p-j, k) : j < k \leq q\}.$$

These chains are disjoint and cover the rectangle. The first and last ranks of the j th chain are j and $p+q-j$. Thus each is symmetric. If the original chain starts at an offset rank, the same construction preserves symmetry about the rank of the full product. Induction proves the decomposition. Every symmetric chain meets a middle rank exactly once, and every antichain meets a chain at most once. This proves the last assertion. This is the symmetric-chain construction of de Bruijn, van Ebbenhorst Tengbergen, and Kruyswijk [4]. $\square$

Proposition 3.2 (Even-moment comparison). *If $B \subset \mathbb{Z}$ has n elements and $r \geq 1$, then*

$$\int_{\mathbb{T}} |F_B|^{2r} \leq [z^{r(n-1)}](1 + z + \cdots + z^{n-1})^{2r} = \int_{\mathbb{T}} |F_{\{0, \dots, n-1\}}|^{2r}. \quad (3.1)$$

In particular

$$E(B) \leq \frac{2n^3 + n}{3}. \quad (3.2)$$

Proof. Write $B = \{b_0 < \cdots < b_{n-1}\}$. An additive relation counted by the left side of (3.1) is represented by a point $x \in \{0, \dots, n-1\}^{2r}$ satisfying

$$b_{x_1} + \cdots + b_{x_r} - b_{n-1-x_{r+1}} - \cdots - b_{n-1-x_{2r}} = 0.$$

The expression on the left strictly increases when any coordinate increases. Its zero set is therefore an antichain. By Lemma 3.1, it has size at most the central rank of the grid, which is the coefficient in (3.1). For the interval, every point of that rank satisfies the required equality, so equality holds. For $r = 2$, the interval difference representation function is $n - |h|$ for $|h| < n$. Its squared sum is $n^2 + 2 \sum_{j=1}^{n-1} j^2 = (2n^3 + n)/3$. $\square$

Corollary 3.3 (Uniform normalized moments). *Let $u_B = |F_B|/n$ and let c_k be as in (1.4). For each fixed integer $k \geq 2$,*

$$n \int_{\mathbb{T}} u_B^k \leq c_k + o_k(1) \quad (n \rightarrow \infty), \quad (3.3)$$

uniformly in B . For every $k \geq 2$ and every B , the left side is at most one. For $r \geq 1$,

$$I_{2r} = \frac{1}{(2r-1)!} \sum_{j=0}^r (-1)^j \binom{2r}{j} (r-j)^{2r-1}. \quad (3.4)$$

Proof. For the interval polynomial $D_n(t) = \sum_{j=0}^{n-1} e(jt)$, change variables $x = nt$ on $[-1/2, 1/2]$. The rescaled integrand is

$$\left| \frac{\sin \pi x}{n \sin(\pi x/n)} \right|^{2r} \mathbf{1}_{[-n/2, n/2]}(x).$$

It converges pointwise to $|\sin(\pi x)/(\pi x)|^{2r}$ and is bounded by a constant times $\min(1, |x|^{-2r})$. Dominated convergence and Proposition 3.2 prove (3.3) for even k . Cauchy–Schwarz between the adjacent even moments gives the odd case. The universal bound follows from $0 \leq u_B \leq 1$ and $n \int u_B^2 = 1$.

For completeness, the coefficient on the right of (3.1) equals

$$\sum_{j=0}^{r-1} (-1)^j \binom{2r}{j} \binom{(r-j)n + r - 1}{2r - 1},$$

where binomial coefficients with too small a top entry are interpreted as zero. Divide by n^{2r-1} and let $n \rightarrow \infty$. Each term has the limit $(r-j)^{2r-1}/(2r-1)!$. The additional $j = r$ term in (3.4) vanishes. This proves the formula and also shows directly that the limiting moment constants are rational. $\square$

4. OUTER MULTIPLIERS AND THE IMPROVED CONSTANT

4.1. An exact pointwise budget. We work with multipliers whose Fourier support is nonpositive. This orientation is convenient for increasing initial segments. We use standard Hardy-space boundary convergence as in Duren [5]. The following construction is the boundary-value form of an outer function, reflected from the usual analytic convention.

Lemma 4.1 (Outer damping). *Let $0 < a < 1$, and let $u : \mathbb{T} \rightarrow [0, 1]$ be measurable. There is an L^∞ function M with*

$$\text{supp } \widehat{M} \subseteq \mathbb{Z}_{\leq 0}, \quad |M| = 1 - au \text{ a.e.}, \quad \widehat{M}(0) = \exp\left(\int_{\mathbb{T}} \log(1 - au)\right) > 0. \quad (4.1)$$

If $M_1, \dots, M_v$ are such multipliers and $P = \prod_j M_j$, then

$$\int_{\mathbb{T}} P = \prod_j \widehat{M}_j(0). \quad (4.2)$$

Proof. Put $\ell = \log(1 - au)$, a real bounded function. For $0 < \rho < 1$ define

$$M_\rho(t) = \exp\left(\widehat{\ell}(0) + 2 \sum_{k < 0} \widehat{\ell}(k) \rho^{|k|} e(kt)\right).$$

The series converges absolutely for each ρ . Its real part before exponentiation is the Poisson integral of ℓ , so $1 - a \leq |M_\rho| \leq 1$. These are the radial values of a bounded analytic function in the reflected disc. The radial boundary function M exists almost everywhere and in every finite L^p . Poisson convergence gives $|M| = e^\ell = 1 - au$, while its constant coefficient is $e^{\widehat{\ell}(0)}$. This proves (4.1). Products are again bounded reflected-analytic functions, and their values at the origin multiply. Equivalently, radial approximation followed by L^1 convergence proves (4.2). $\square$

For a finite nonempty B , let $M_{a,B}$ denote this multiplier for $u_B = |F_B|/|B|$. Whenever $|G| \leq 1$, the update

$$G \mapsto GM_{a,B} + aF_B/|B| \quad (4.3)$$

preserves the bound one, because the two summands have total modulus at most $(1 - au_B) + au_B = 1$. Unlike an exponential majorant for the modulus, this update leaves no unused pointwise room.

4.2. Ordered pairing and the whole-product error. Write $A = \{a_1 < \dots < a_N\}$ and let A_i be increasing initial segments of sizes n_i . For now the number of segments is arbitrary. Set $M_i = M_{a,A_i}$ and

$$P_i = \prod_{j > i} M_j, \quad G = a \sum_i \frac{F_{A_i}}{n_i} P_i. \quad (4.4)$$

Iterating (4.3) from zero proves $\|G\|_\infty \leq 1$. The frequency orientation gives the following exact localization of the pairing.

Lemma 4.2 (The initial-segment pairing). *For every initial segment A_i and every P with nonpositive Fourier support,*

$$|\langle F_{A_i}, (F_{A_i}/n_i)P \rangle - 1| \leq \left(\frac{E(A_i) + n_i^2}{2n_i^2}\right)^{1/2} \|P - 1\|_2. \quad (4.5)$$

Here P may be any L^2 function for which the displayed pairing is formed; in particular it may be the bounded product P_i .

Proof. Let $p_s = \widehat{P}(s)$. In the pairing, a coefficient p_s can occur only when $a = b + s$, with $a \in A$, $b \in A_i$, and $s \leq 0$. Then $a \leq b$, so the initial-segment property implies $a \in A_i$. Consequently the pairing is unchanged if F_A is replaced by F_{A_i} . For $s \in \mathbb{Z}$ write

$$d_s = \frac{1}{n_i} \#\{(a, b) \in A_i^2 : a - b = s\}.$$

Then $d_0 = 1$, $d_s = d_{-s}$, and $\sum_s d_s^2 = E(A_i)/n_i^2$. Subtracting the contribution of the constant function $P = 1$, Cauchy–Schwarz on $s \leq 0$ gives

$$\left| \sum_{s \leq 0} d_s \overline{(p_s - \mathbf{1}_{s=0})} \right| \leq \left(\sum_{s \leq 0} d_s^2 \right)^{1/2} \|P - 1\|_2.$$

The sum of squares in the last expression is $(E(A_i)/n_i^2 + 1)/2$, as claimed. All sums involving d_s are finite, so no issue of pointwise Fourier convergence arises. $\square$

Define, for $0 \leq x \leq 1$,

$$h_a(x) = a^2 x^2 - 2ax - 2 \log(1 - ax), \quad q_a(x) = 2ax - a^2 x^2. \quad (4.6)$$

The function h_a has the positive power series

$$h_a(x) = 2a^2 x^2 + 2 \sum_{k \geq 3} \frac{a^k x^k}{k}. \quad (4.7)$$

Lemma 4.3 (Whole-product estimate). *Let M_j be the multipliers in Lemma 4.1, associated to $u_j : \mathbb{T} \rightarrow [0, 1]$, and put $P = \prod_j M_j$. Then*

$$\|P - 1\|_2^2 \leq \sum_j \int_{\mathbb{T}} h_a(u_j) + \sum_{j < k} \int_{\mathbb{T}} q_a(u_j) q_a(u_k). \quad (4.8)$$

Proof. Set

$$W = \prod_j (1 - au_j), \quad S = \sum_j \int_{\mathbb{T}} -\log(1 - au_j).$$

By (4.2), $\int P = e^{-S}$, a positive real number. Hence

$$\|P - 1\|_2^2 = \int_{\mathbb{T}} W^2 + 1 - 2e^{-S}. \quad (4.9)$$

We have $W^2 = \prod_j (1 - q_a(u_j))$ and $0 \leq q_a(u_j) \leq 1$. The second Bonferroni inequality gives the pointwise upper bound

$$W^2 \leq 1 - \sum_j q_a(u_j) + \sum_{j < k} q_a(u_j) q_a(u_k).$$

For instance, this is the usual two-term bound for the probability that none of independent events of probabilities $q_a(u_j)$ occurs, applied at a fixed point of $\mathbb{T}$. Since $1 - e^{-S} \leq S$, substitution in (4.9) proves (4.8), using $-q_a(x) - 2 \log(1 - ax) = h_a(x)$. $\square$

The point of (4.8) is not merely the retention of a cross term. The quantities $h_a(u)$ and $q_a(u)^2$ depend on the same variable. They cannot in general be simultaneously as large as their separate uniform upper bounds.

4.3. A coupled scalar majorant. We now fix $a = 9/10$ and impose the exact scale ratio $n_{i+r} = 9^r n_i$. For a fixed i , let

$$D_r = n_{i+r} \int_{\mathbb{T}} h_a(u_{A_{i+r}}), \quad Q_r = n_{i+r} \int_{\mathbb{T}} q_a(u_{A_{i+r}})^2, \quad w_r = 3^{-r}.$$

Cauchy–Schwarz on each cross term of (4.8) gives

$$n_i \|P_i - 1\|_2^2 \leq \sum_r w_r^2 D_r + \sum_{r < s} w_r w_s \sqrt{Q_r Q_s}. \quad (4.10)$$

All indices here range only over the existing tail of the chain. Choose the infinite sequence

$$v_1 = \frac{1}{4}, \quad v_r = 3^{-r} \quad (r \geq 2), \quad V_0 = \sum_{r \geq 1} v_r = \frac{5}{12}, \quad b_r = \frac{1}{2} \left(\frac{V_0}{v_r} - 1 \right). \quad (4.11)$$

In particular $b_1 = 1/3$ and $b_r \geq b_2 = 11/8$ for $r \geq 2$. Weighted Cauchy–Schwarz, with the possibly unused v_r retained in V_0 , gives

$$\left(\sum_r w_r \sqrt{Q_r} \right)^2 \leq V_0 \sum_r \frac{w_r^2 Q_r}{v_r}.$$

It follows from (4.10) that

$$n_i \|P_i - 1\|_2^2 \leq \sum_r w_r^2 (D_r + b_r Q_r). \quad (4.12)$$

Lemma 4.4 (Scalar coupling). *Let $a = 9/10$. For a nonempty finite integer set B , $n = |B|$, put $u = |F_B|/n$. If $b \geq 11/8$, then*

$$n \int_{\mathbb{T}} (h_a(u) + b q_a(u)^2) \leq (2 + 4b) a^2. \quad (4.13)$$

For $b = 1/3$, uniformly as $n \rightarrow \infty$,

$$n \int_{\mathbb{T}} (h_a(u) + \frac{1}{3} q_a(u)^2) \leq \frac{10a^2}{3} + \Delta_a + o(1), \quad (4.14)$$

where

$$\Delta_a = \frac{2}{3} \left(\frac{5a^4}{6} - \frac{2a^3}{3} \right) + 2 \sum_{k \geq 5} \frac{c_k a^k}{k}. \quad (4.15)$$

Proof. By Parseval, $d\nu = nu^2 dt$ is a probability measure. Define

$$\Phi_b(x) = \frac{h_a(x)}{x^2} + b(2a - a^2 x)^2 \quad (x > 0), \quad \Phi_b(0) = (2 + 4b)a^2.$$

The left sides under consideration equal $\int \Phi_b(u) d\nu$. The positive series (4.7), together with the convex quadratic in this definition, shows that Φ_b is convex on $[0, 1]$ for every $b \geq 0$. Its endpoint difference is

$$\Phi_b(1) - \Phi_b(0) = -a^2 - 2a - 2\log(1 - a) - b(4a^3 - a^4). \quad (4.16)$$

For $a = 9/10$ and $b \geq 11/8$ this is negative. This can be checked with rational arithmetic and $\log 10 < 5/2$; a sharper enclosure is recorded in Appendix A. Convexity therefore gives $\Phi_b(x) \leq \Phi_b(0)$ throughout $[0, 1]$, proving (4.13).

For $b = 1/3$, expansion gives

$$\Phi_{1/3}(x) = \frac{10a^2}{3} - \frac{2a^3}{3}x + \frac{5a^4}{6}x^2 + 2 \sum_{k \geq 5} \frac{a^k x^{k-2}}{k}.$$

As $x \geq x^2$ on $[0, 1]$ and $5a^4/6 - 2a^3/3 > 0$ for our value of a ,

$$\Phi_{1/3}(x) \leq \frac{10a^2}{3} + \left(\frac{5a^4}{6} - \frac{2a^3}{3} \right) x^2 + 2 \sum_{k \geq 5} \frac{a^k x^{k-2}}{k}.$$

Integrating against ν and applying Corollary 3.3, with $c_4 = I_4 = 2/3$, proves (4.14). To justify the infinite sum uniformly in B , first truncate at a fixed index. The remaining contribution is at most $2 \sum_{k > L} a^k/k$, since $n \int u^k \leq 1$. This tends to zero independently of B , after which the finite number of moment limits may be taken. $\square$

Proposition 4.5 (Uniform tail budget). *For the ratio-nine chain above, as its smallest size tends to infinity,*

$$n_i \|P_i - 1\|_2^2 \leq \mathcal{E}_{9/10} + o(1) \quad (4.17)$$

uniformly in i , in the length of the chain, and in the underlying set.

Proof. Apply Lemma 4.4 in (4.12). All terms with $r \geq 2$ are bounded without error, and only $r = 1$ uses a moment limit. If there is no tail, the result is immediate. Otherwise the required moment error is uniform once n_{i+1} tends to infinity. The constant part is bounded by the infinite sum

$$a^2 \sum_{r \geq 1} w_r^2 (2 + 4b_r) = 2a^2 V_0 \sum_{r \geq 1} \frac{w_r^2}{v_r} = 2a^2 \frac{5}{12} \left(\frac{4}{9} + \frac{1}{6} \right) = \frac{55a^2}{108}.$$

The first-level correction is $w_1^2 \Delta_a = \Delta_a/9$. This is exactly $\mathcal{E}_a$ from (1.5). Adding unused nonnegative terms causes no difficulty for a finite tail. $\square$

4.4. Completion of the quantitative theorem.

Proof of Theorem 1.1. For large N , put $n_1 = \lceil \log N \rceil$ and $n_i = 9^{i-1} n_1$ for $1 \leq i \leq J$, where J is maximal with $n_J \leq N$. Thus

$$J = \frac{\log N - \log \log N}{\log 9} + O(1). \quad (4.18)$$

Let A_i be the first n_i elements of A , and use the bounded test G in (4.4). The interval energy bound (3.2) gives

$$\frac{E(A_i) + n_i^2}{2n_i^2} \leq \frac{n_i}{3} + \frac{1}{2} + \frac{1}{6n_i}.$$

Together with Lemma 4.2 and Proposition 4.5, this implies, uniformly in i ,

$$\operatorname{Re} \langle F_A, (F_{A_i}/n_i) P_i \rangle \geq 1 - \sqrt{\mathcal{E}_{9/10}/3} - o(1).$$

Because $\|G\|_\infty \leq 1$, summing and using (4.18) yields

$$\mathcal{L}(A) \geq \operatorname{Re} \langle F_A, G \rangle \geq \left[\frac{9}{10 \log 9} \left(1 - \sqrt{\mathcal{E}_{9/10}/3} \right) - o(1) \right] \log N.$$

Every error used here depends only on N , not on the diameter or arithmetic arrangement of A . Finally, Appendix A proves with rational bounds that

$$\mathcal{E}_{9/10} < 0.479084457020255861, \quad c_* > 0.2459209213577253.$$

In particular $c_* > 0.2459209$, as stated. $\square$

Remark 4.6. The last theorem is an asymptotic leading-coefficient statement. It does not assert the same decimal times $\log N$ with no error for every small N , nor does it optimize the entire class of outer recursions. Its frequency support still has a one-sided endpoint restriction. The classical Landau bound in Appendix B explains why optimizing only within that final spectral class cannot reach $4/\pi^2$.

5. HEREDITARY ENERGY AND REPEATED STRUCTURAL EXTRACTION

The structural proof needs an invariant that survives deletion. A bound for the Fourier norm of an original set need not survive when a subset is removed. An energy inequality formulated for *every* remaining subset does survive, and this is the purpose of the next lemma.

Lemma 5.1 (Hereditary energy gives an almost-covering). *Let X be a nonempty finite subset of an abelian group, $|X| = R$, and suppose that, for some $0 < \eta \leq 1$,*

$$E(Y) \geq \eta \frac{|Y|^4}{R} \quad (\emptyset \neq Y \subseteq X). \quad (5.1)$$

For every $0 < \delta < 1$, there are pairwise disjoint subsets $X_1, \dots, X_J \subseteq X$ such that

$$\left| X \setminus \bigcup_{j=1}^J X_j \right| < \delta R, \quad J \leq C\eta^{-C}\delta^{-C}, \quad (5.2)$$

$$|X_j| \geq c\eta^C\delta^C R, \quad |X_j + X_j| \leq C\eta^{-C}\delta^{-C}|X_j|. \quad (5.3)$$

All constants here are absolute.

Proof. Take the polynomial constants from Theorem 2.2. If the current remainder Y has $|Y| \geq \delta R$, then (5.1) gives $E(Y) \geq \eta\delta|Y|^3$. BSG produces $Z \subseteq Y$ of size at least $c(\eta\delta)^C|Y|$, and with doubling at most $C(\eta\delta)^{-C}$. In particular the number removed is at least $c(\eta\delta)^C\delta R$. Remove this Z and repeat. The selected sets are disjoint, every step removes a fixed parameter-dependent fraction of the *original* size R , and the process stops after a bounded number of steps. Increasing the absolute exponent absorbs the additional factor of δ in the size and step bounds. The only hypothesis used at a later step is (5.1) for its remainder. $\square$

Lemma 5.2 (A constant spectral budget gives hereditary energy). *Let X be a nonempty finite subset of an abelian group, let $R = |X|$, and put $S = \|F_X\|_1$ on its compact dual. For every $Y \subseteq X$,*

$$E(Y) \geq \frac{|Y|^4}{S^2 R}. \quad (5.4)$$

Proof. The claim is immediate for $Y = \emptyset$. Otherwise $\langle F_X, F_Y \rangle = |Y|$. Weighted Cauchy–Schwarz and then ordinary Cauchy–Schwarz imply

$$\begin{aligned} |Y|^2 &\leq \left(\int |F_X| |F_Y| \right)^2 \\ &\leq \left(\int |F_X| \right) \left(\int |F_X| |F_Y|^2 \right) \\ &\leq S \left(\int |F_X|^2 \right)^{1/2} \left(\int |F_Y|^4 \right)^{1/2} = S\sqrt{R}\sqrt{E(Y)}. \end{aligned}$$

Squaring and rearranging proves the result. In particular, this argument uses no order on the group and applies on a finite cyclic group. $\square$

Applied directly with $S = K \log N$, the last lemma still loses powers of $\log N$. Its role below is to act *after* a long integer direction has been removed, when the residual spectral budget is a constant. For separated blocks, we instead obtain (5.1) from bounded nonlinear amplification.

6. BOUNDED AMPLIFICATION AND THE ENERGY OF INTEGER BLOCK LABELS

6.1. A positive density of balanced correlations.

Lemma 6.1 (Dense mixed correlations). *Let $F \in L^1(\mathbb{T})$, $L = \|F\|_1 > 0$, and let $\Phi_1, \dots, \Phi_r$ satisfy*

$$\|\Phi_i\|_\infty \leq 1, \quad \operatorname{Re}\langle F, \Phi_i \rangle \geq \kappa > 0.$$

Put $T = L/\kappa \geq 1$, and let m be the smallest odd integer at least $64T^2$. If $r \geq m$, then there is an odd integer $k = 2s - 1$, $3 \leq k \leq m$, such that at least $e^{-CT^2}r^k$ ordered k -tuples of distinct indices satisfy

$$\left| \langle F, \Phi_{i_1} \cdots \Phi_{i_s} \overline{\Phi_{i_{s+1}} \cdots \Phi_{i_k}} \rangle \right| \geq e^{-CT^2} \kappa. \quad (6.1)$$

No arithmetic assumption on the test functions is made.

Proof. For odd m , let $\text{Maj}_m : \{-1, 1\}^m \rightarrow \{-1, 1\}$ be the sign of the sum of its coordinates, and consider its multi-affine extension

$$P_m(t_1, \dots, t_m) = \sum_{S \subseteq [m]} \widehat{\text{Maj}_m}(S) \prod_{j \in S} t_j.$$

This extension is bounded in absolute value by one throughout $[-1, 1]^m$: it is the expectation of Boolean majority under independent signs whose expectations are the prescribed t_j . Only odd degrees occur. Each first-degree coefficient equals

$$a_m = 2^{-(m-1)} \binom{m-1}{(m-1)/2} \geq (2m)^{-1/2}.$$

Indeed the other $m-1$ signs contribute to the correlation with one coordinate exactly when they sum to zero. The displayed lower bound follows from the elementary central-binomial estimate. Boolean Parseval and Cauchy–Schwarz also give

$$B_m := \sum_{S \subseteq [m]} |\widehat{\text{Maj}_m}(S)| \leq 2^{m/2}. \quad (6.2)$$

To handle complex functions without losing the balance of conjugates, work on $\mathbb{T}^2$ and put

$$H(x, t) = 2 \operatorname{Re}(e(t)F(x)), \quad \psi_i(x, t) = \operatorname{Re}(e(t)\Phi_i(x)).$$

Then $|\psi_i| \leq 1$ and direct integration in t gives

$$\|H\|_1 = \frac{4}{\pi} L =: L_*, \quad \langle H, \psi_i \rangle = \operatorname{Re} \langle F, \Phi_i \rangle \geq \kappa. \quad (6.3)$$

For any injection $\iota : [m] \rightarrow [r]$, pair H with $P_m(\psi_{\iota(1)}, \dots, \psi_{\iota(m)})$. The full pairing has absolute value at most L_* . Its linear contribution is at least $A_m \kappa$, where

$$A_m = m a_m \geq \sqrt{m/2} \geq 2L_*/\kappa.$$

Average over all injections. For an odd degree k , let

$$J_k = \mathbb{E}_{i_1, \dots, i_k \text{ distinct}} \langle H, \psi_{i_1} \cdots \psi_{i_k} \rangle.$$

Every monomial of degree k has the same averaged pairing J_k . The higher terms must offset at least $A_m \kappa - L_* \geq A_m \kappa/2$. By (6.2), some odd $k \geq 3$ therefore satisfies

$$|J_k| \geq \frac{A_m}{2B_m} \kappa =: \tau \kappa, \quad \tau \geq e^{-Cm}. \quad (6.4)$$

Write $k = 2s - 1$. Expanding the real parts and integrating in the extra phase yields the exact identity

$$\langle H, \prod_{j=1}^k \psi_{i_j} \rangle = 2^{1-k} \operatorname{Re} \sum_{\substack{U \subseteq [k] \\ |U|=s}} \langle F, \prod_{j \in U} \Phi_{i_j} \prod_{j \notin U} \overline{\Phi_{i_j}} \rangle. \quad (6.5)$$

Only products with one more unconjugated factor survive. After averaging, all the terms on the right agree by permutation symmetry. Since $2^{1-k} \binom{k}{s} \leq 1$, (6.4) implies

$$\mathbb{E}_{\text{distinct}} \left| \langle F, \Phi_{i_1} \cdots \Phi_{i_s} \overline{\Phi_{i_{s+1}}} \cdots \overline{\Phi_{i_k}} \rangle \right| \geq \tau \kappa.$$

Each absolute pairing is at most $L = T\kappa$. Hence at least a $\tau/(2T)$ fraction of the distinct tuples have absolute pairing at least $\tau\kappa/2$. Finally,

$$\frac{(r)_k}{r^k} \geq \frac{k!}{k^k} \geq e^{-k} \quad (r \geq k),$$

where $(r)_k = r(r-1) \cdots (r-k+1)$. Since $k \leq m \ll T^2$, all fixed factors and the factor T can be absorbed into e^{CT^2} . This proves (6.1) with an absolute C . $\square$

The distinction between a single nonzero mixed correlation and a positive density of such correlations is essential. The latter can be counted as additive relations. The argument has also retained all higher terms of the bounded polynomial; it has not presumed that a cubic correction alone captures the obstruction.

6.2. Finite-band certificates for actual integer blocks. Let C have the exact representation (1.8), initially without the density conditions in (1.9). We require only $D \geq 1$, $C_i \neq \emptyset$, and separation $|b_i - b_j| > 4D$.

Lemma 6.2 (A finite-band certificate). *For every $B \subseteq [-D, D] \cap \mathbb{Z}$ there is a trigonometric polynomial ψ_B such that*

$$\|\psi_B\|_\infty \leq 1, \quad \text{supp } \widehat{\psi_B} \subseteq [-2D, 2D], \quad \langle F_B, \psi_B \rangle = \frac{1}{3} \mathcal{L}(B).$$

Proof. Let $\mathcal{F}_D$ be the Fejér kernel whose coefficients are $(1 - |n|/D)_+$, and put $V_D = 2\mathcal{F}_{2D} - \mathcal{F}_D$. Its coefficients equal one on $[-D, D]$, vanish outside $(-2D, 2D)$, and $\|V_D\|_1 \leq 3$. Define the phase of F_B to be zero at its zeros, and put

$$\psi_B = \frac{1}{3} V_D * \frac{F_B}{|F_B|}.$$

Convolution gives the L^∞ and support assertions. Since V_D is real and even and has coefficient one at every frequency of B ,

$$\langle F_B, \psi_B \rangle = \frac{1}{3} \langle V_D * F_B, F_B / |F_B| \rangle = \frac{1}{3} \int |F_B|.$$

The empty-set case is covered by the zero function. $\square$

Take $\Phi_i = e(b_i t) \psi_{C_i}$. Its support lies in $b_i + [-2D, 2D]$. The separation of the centres implies that no other block $b_j + C_j$ meets this support. Consequently

$$\langle F_C, \Phi_i \rangle = \frac{1}{3} \mathcal{L}(C_i). \quad (6.6)$$

Let

$$\ell_i = \lfloor b_i/D \rfloor, \quad \Lambda = \{\ell_i : 1 \leq i \leq R\}, \quad \kappa = \frac{1}{3} \min_i \mathcal{L}(C_i), \quad T = \mathcal{L}(C)/\kappa. \quad (6.7)$$

The labels are distinct and $T \geq 1$ by (6.6).

Proposition 6.3 (Hereditary energy of the labels). *For every nonempty $S \subseteq \Lambda$,*

$$E(S) \geq e^{-CT^2} \frac{|S|^4}{R}. \quad (6.8)$$

The parameter T refers to the full original target C and is the same for every S .

Proof. Write $r = |S|$, and let m be as in Lemma 6.1. If $r < m$, then $E(S) \geq r^2$ suffices: because $R \geq r$ and $m \ll T^2$, an absolute enlargement of C ensures $e^{-CT^2} r^2 \leq R$. Assume therefore that $r \geq m$. Apply Lemma 6.1 to the certificates indexed by S , keeping F_C as target. For some $k = 2s - 1 \ll T^2$, at least $e^{-CT^2} r^{2s-1}$ ordered distinct tuples have nonzero balanced correlation.

The Fourier support of such a balanced product lies in

$$b_{i_1} + \cdots + b_{i_s} - b_{i_{s+1}} - \cdots - b_{i_k} + [-2kD, 2kD].$$

Nonzero correlation with F_C forces this interval to contain a point of one of the original blocks. Hence, for some $j \in \{1, \dots, R\}$,

$$|b_{i_1} + \cdots + b_{i_s} - b_{i_{s+1}} - \cdots - b_{i_k} - b_j| \leq (2k + 1)D.$$

The witness j is not required to have $\ell_j \in S$. Writing $b_i = D\ell_i + r_i$, $0 \leq r_i < D$, gives an integer t with

$$\ell_{i_1} + \cdots + \ell_{i_s} - \ell_{i_{s+1}} - \cdots - \ell_{i_k} - \ell_j = t, \quad |t| \leq 4k. \quad (6.9)$$

Indeed the rounding error contains $k+1$ remainders, so $(2k+1) + (k+1)$ provides a sufficient bound for $k \geq 3$. Choose one witness for each good tuple. Different tuples still give different ordered solutions when the witness is included. With $r_s = \mathbf{1}_S^*$, their count yields

$$\sum_{|t| \leq 4k} \sum_x r_s(x) (\mathbf{1}_S^{*(s-1)} * \mathbf{1}_\Lambda)(x+t) \geq e^{-CT^2} r^{2s-1}. \quad (6.10)$$

Changing t to $-t$ if necessary gives exactly this convention.

By Fourier inversion and then Hölder, each inner shifted pairing is bounded above by

$$\begin{aligned} \int_{\mathbb{T}} |F_S|^{2s-1} |F_\Lambda| &\leq r^{2s-4} \int_{\mathbb{T}} |F_S|^3 |F_\Lambda| \\ &\leq r^{2s-4} E(S)^{3/4} E(\Lambda)^{1/4} \\ &\leq r^{2s-4} E(S)^{3/4} R^{3/4}. \end{aligned}$$

Here $s \geq 2$, and the last inequality uses the elementary $E(\Lambda) \leq R^3$. Combining with (6.10), absorbing $8k+1$ into the exponential constant, and taking the power $4/3$ proves (6.8). $\square$

In particular, the high-order correlations have been reduced to ordinary fourth-order energy. No assertion that the floor map is a Freiman isomorphism is used: the bounded integer error in (6.9) is counted rather than discarded.

Theorem 6.4 (Almost-covering for dense separated blocks). *Fix $K_0 \geq 1$, $0 < \beta \leq 1$, $\Delta \geq 1$, and $0 < \varepsilon < 1$. If a set C satisfies (1.8)–(1.9), $|C| = n \geq 2$, and $\mathcal{L}(C) \leq K_0 \log n$, then all but at most εn points of C are covered by boundedly many pairwise disjoint subsets, each of size at least cn and doubling at most C_1 . The positive constants and the number of subsets depend only on $K_0, \beta, \Delta, \varepsilon$.*

Proof. The weighted MPS inequality gives

$$\kappa \geq c \log(M+1) \geq c\beta \log n, \quad T \leq CK_0/\beta.$$

Apply Lemma 5.1 to (6.8), with $\delta = \varepsilon/(6\Delta)$. The omitted labels number less than δR . Since $|C_i| \leq 2D+1 \leq 3\Delta M$ and $n \geq MR$, their original blocks contain at most

$$(2D+1)\delta R \leq 3\Delta\delta n \leq \varepsilon n/2$$

points. For each selected label set S_j , take B_j to be the union of all its original blocks. The B_j are disjoint and $|B_j| \geq M|S_j|$. Also $n \leq 3\Delta MR$, so the lower size bound for S_j gives $|B_j| \geq cn$.

If $|S_j + S_j| \leq Q|S_j|$, each point in a corresponding integer block has the form $D\ell + u$ with $u \in [-D, 2D] \cap \mathbb{Z}$. Thus

$$|B_j + B_j| \leq (6D+1)|S_j + S_j| \leq 7\Delta Q|B_j|.$$

The label constants are bounded in the stated parameters, and so are all the constants in the conclusion. In particular no estimate for $\mathcal{L}$ of a deleted-block remainder was required. $\square$

7. INTEGER BOUNDARIES AND LONG ARITHMETIC BLOCKS

The second local argument converts a logarithmic budget into a constant spectral budget. The relevant object is an integer-valued boundary on a cyclic group times $\mathbb{Z}$, rather than a collection of independently chosen analytic vectors.

7.1. Extracting the boundary norm. Let $q, M \geq 1$ and let $\emptyset \neq \mathcal{C} \subset (\mathbb{Z}/q\mathbb{Z}) \times \mathbb{Z}$ be finite, of size V . Initially no aspect condition is imposed. Residues b are always represented by $0 \leq b < q$. Put $A = A_M(\mathcal{C})$ as in (1.10), and define

$$h(b, u) = \mathbf{1}_{\mathcal{C}}(b, u) - \mathbf{1}_{\mathcal{C}}(b, u - 1), \quad H = \sum_{b, u} |h(b, u)|^2 \leq 2V, \quad (7.1)$$

$$\mathcal{H}(j, s) = \sum_{b, u} h(b, u) e(bj/q + us), \quad S = \frac{1}{q} \sum_{j=0}^{q-1} \int_{\mathbb{T}} |\mathcal{H}(j, s)| ds. \quad (7.2)$$

The function h is nonzero, takes values in $\{-1, 0, 1\}$, and has zero sum in u for every b . Fourier inversion and Parseval give

$$1 \leq S \leq \sqrt{H}. \quad (7.3)$$

Proposition 7.1 (Uniform boundary extraction). *For every $q, M, \mathcal{C}$ as above,*

$$\mathcal{L}(A_M(\mathcal{C})) \geq \frac{S}{\pi} \log \frac{MS}{\sqrt{H}} - 4S. \quad (7.4)$$

For fixed $q, \mathcal{C}$, as $M \rightarrow \infty$ one also has

$$\mathcal{L}(A_M(\mathcal{C})) = \frac{S}{\pi} \log M + O_{q, \mathcal{C}}(1). \quad (7.5)$$

Proof. The finite geometric series for each cell gives the exact identity

$$(1 - e(q\theta))F_A(\theta) = \sum_{b, u} h(b, u) e(b\theta + qMu\theta). \quad (7.6)$$

Partition the circle by writing $\theta = (j + t)/q$, with $0 \leq j < q$ and $-1/2 \leq t < 1/2$. Define

$$W(t, s) = \frac{1}{q} \sum_{j=0}^{q-1} \left| \sum_{b, u} h(b, u) e(b(j + t)/q + us) \right|.$$

Equation (7.6) yields

$$\mathcal{L}(A) = \int_{-1/2}^{1/2} \frac{W(t, Mt)}{2|\sin \pi t|} dt. \quad (7.7)$$

The values at the isolated zeros of the denominator are interpreted by the original identity and do not affect the integral.

Let

$$W_0(s) = W(0, s), \quad Q(s) = \left(\sum_b \left| \sum_u h(b, u) e(us) \right|^2 \right)^{1/2}.$$

For fixed s , the difference inside $W(t, s)$ and $W_0(s)$ is a Fourier sum in j with coefficients $(e(bt/q) - 1) \sum_u h(b, u) e(us)$. The triangle inequality, discrete Parseval, and $|e(bt/q) - 1| \leq 2\pi|t|$ prove

$$|W(t, s) - W_0(s)| \leq 2\pi|t|Q(s). \quad (7.8)$$

Since h is real, W_0, Q are even and one-periodic, and $W(-t, -s) = W(t, s)$ after conjugation and reindexing j . Furthermore

$$\int_0^1 W_0(s) ds = S, \quad \int_0^1 Q(s) ds \leq \sqrt{H}. \quad (7.9)$$

The second assertion follows from Cauchy-Schwarz and Parseval in s .

Restrict (7.7) to $|t| \leq a$, where $0 < a \leq 1/2$. Since $2t \leq \sin \pi t \leq \pi t$ for $0 \leq t \leq 1/2$, (7.8) gives

$$\mathcal{L}(A) \geq \frac{1}{\pi} \int_0^{Ma} \frac{W_0(u)}{u} du - \pi(a + M^{-1})\sqrt{H}. \quad (7.10)$$

For the error term, split $[0, Ma]$ into complete unit periods and one remaining interval; nonnegativity and periodicity bound its integral of Q by $(Ma + 1) \int_0^1 Q$.

A nonnegative one-periodic function of mean S satisfies

$$\int_0^X \frac{W_0(u)}{u} du \geq S(\log X - 2) \quad (X \geq 1). \quad (7.11)$$

To see this, discard $[0, 1]$ and the final incomplete interval. Each complete interval $[j, j + 1]$, $j \geq 1$, contributes at least $S/(j + 1)$. The elementary harmonic-sum bound proves (7.11). Now choose $a = S/(4\sqrt{H})$, which lies in $(0, 1/4]$ by (7.3). If $MS/\sqrt{H} \geq 4$, then $Ma \geq 1$ and the error in (7.10) is at most $\pi S/2$. Hence

$$\mathcal{L}(A) \geq \frac{S}{\pi} \log \frac{MS}{\sqrt{H}} - \left(\frac{\log 4 + 2}{\pi} + \frac{\pi}{2} \right) S,$$

which is stronger than (7.4). If $MS/\sqrt{H} < 4$, the right side of (7.4) is negative, so that inequality is automatic.

For (7.5), hold $q, \mathcal{C}$ fixed. On the whole interval $[-1/2, 1/2]$, the cruder bound $|W(t, s) - W_0(s)| \leq 2\pi|t| \sum |h|$ has an integrable quotient by $|\sin \pi t|$. Replacing $1/(2|\sin \pi t|)$ by $1/(2\pi|t|)$ also has bounded integrated error. Thus

$$\mathcal{L}(A) = \frac{1}{\pi} \int_0^{M/2} \frac{W_0(u)}{u} du + O_{q, \mathcal{C}}(1).$$

The integral near zero is finite: $W_0(0) = 0$, because each vertical sum of h is zero, and W_0 is Lipschitz. The periodic function $W_0 - S$ has a bounded primitive; integration by parts on $[1, M/2]$ therefore shows that its contribution divided by u is bounded. This proves (7.5). $\square$

Corollary 7.2 (A constant boundary budget). *Suppose $M \geq V$, $n = MV$ is sufficiently large absolutely, and $\mathcal{L}(A_M(\mathcal{C})) \leq K_0 \log n$, with $K_0 \geq 1$. Then $S \leq CK_0$. More generally, $M \geq V^{1/2+\sigma}$ with fixed $0 < \sigma \leq 1/2$ gives $S \leq CK_0/\sigma$ once n is sufficiently large in terms of σ .*

Proof. Since $H \leq 2V$ and $S \geq 1$, the first aspect condition gives

$$\log \frac{MS}{\sqrt{H}} \geq \frac{1}{2} \log M - \frac{1}{2} \log 2, \quad \log n \leq 2 \log M.$$

For large M , Proposition 7.1 implies $\mathcal{L}(A_M(\mathcal{C})) \geq (S/(4\pi)) \log M$, proving the first assertion. The condition $M \geq V$ ensures that $M \rightarrow \infty$ as $n \rightarrow \infty$. For the second assertion use

$$\log(M/\sqrt{V}) \geq \frac{2\sigma}{1+2\sigma} \log M, \quad \log n \leq 3 \log M,$$

and absorb the fixed negative multiple of S once more. $\square$

7.2. Integer-valuedness creates shared boundary heights.

Lemma 7.3 (Common heights and residual spectra). *Let $u_1 < \dots < u_v$ be the heights at which $h(\cdot, u)$ is nonzero. Then*

$$v \leq e^{CS}. \quad (7.12)$$

Between consecutive heights, $\mathcal{C}$ is a slab $X_j \times [u_j, u_{j+1})$, with possibly empty $X_j \subseteq \mathbb{Z}/q\mathbb{Z}$. The nonempty slabs partition $\mathcal{C}$, and

$$\|F_{X_j}\|_{L^1(\widehat{\mathbb{Z}/q\mathbb{Z}})} \leq vS. \quad (7.13)$$

Proof. For fixed $a \in \{0, \dots, q-1\}$, apply Theorem 2.1 in the vertical variable to

$$\mathcal{H}(a, s) = \sum_{j=1}^v \left(\sum_b h(b, u_j) e(ba/q) \right) e(u_j s).$$

Some displayed coefficients may vanish; the weighted inequality remains valid with the original indices. Averaging over a gives

$$S \geq c_{\text{MPS}} \sum_{j=1}^v \frac{1}{j} \|F_{h(\cdot, u_j)}\|_{L^1(\widehat{\mathbb{Z}/q\mathbb{Z}})}.$$

Each underlying coefficient function is nonzero and integer-valued, so its Fourier algebra norm is at least one by inversion. Hence $S \geq c_{\text{MPS}} \sum_{j \leq v} 1/j$, proving (7.12).

Extraction of the coefficient at height u_j is contractive:

$$\frac{1}{q} \sum_a \left| \int_{\mathbb{T}} \mathcal{H}(a, s) e(-u_j s) ds \right| \leq S.$$

Thus each boundary-height function has finite-group norm at most S . Telescoping the definition of h shows that, on the vertical interval $[u_j, u_{j+1})$, the indicator of $\mathcal{C}$ is the residue indicator

$$\mathbf{1}_{X_j}(b) = \sum_{i \leq j} h(b, u_i).$$

It is zero before the first and after the last height, because $\mathcal{C}$ is finite. This proves the slab partition and, by the triangle inequality, (7.13). $\square$

It is important that the values of h are integers. Arbitrarily small nonzero real coefficients would not impose a cost of at least one at a new height. It is also important that the residue group remains cyclic: replacing it by independent free coordinates would erase carries.

Lemma 7.4 (Lifting doubling with carries). *Let $X \subseteq \{0, \dots, q-1\}$ and $P = X + q\{a, a+1, \dots, a+L-1\}$, where $L \geq 1$. If $|X + X|_{\mathbb{Z}/q\mathbb{Z}} \leq Q|X|$, then*

$$|P + P| \leq 2Q|P|. \quad (7.14)$$

Proof. For a residue sum write $x_1 + x_2 = r + q\kappa$ with $0 \leq r < q$ and $\kappa \in \{0, 1\}$. The sum of two vertical coordinates in $[a, a+L)$ ranges from $2a$ to $2a+2L-2$. Allowing either carry produces at most $2L$ values for that residue. Therefore $|P + P| \leq 2L|X + X|_{\mathbb{Z}/q\mathbb{Z}} \leq 2QL|X| = 2Q|P|$. $\square$

Theorem 7.5 (Almost-covering for long block inflations). *Fix $K_0 \geq 1$ and $0 < \varepsilon < 1$. Let $C = A_M(\mathcal{C})$ with $M \geq V$, $n = MV$ sufficiently large, and $\mathcal{L}(C) \leq K_0 \log n$. Then all but at most εn points of C are covered by boundedly many disjoint subsets, each of size at least cn and doubling at most C_1 . All constants depend only on K_0, ε .*

Proof. Corollary 7.2 and Lemma 7.3 give $S \leq CK_0$, $v \leq e^{CK_0}$, and a partition into at most v nonempty arithmetic slabs

$$P_j = X_j + q\{a_j, \dots, a_j + L_j - 1\}.$$

Here $L_j = M(u_{j+1} - u_j)$, and the finite-group spectral norm of each X_j is at most vS , hence bounded only in terms of K_0 .

Discard slabs of size less than $\varepsilon n/(4v)$, losing at most $\varepsilon n/4$ points. For a remaining residue set X_j , apply Lemma 5.2 and then Lemma 5.1, with $\delta = \varepsilon/4$. This covers all but a δ proportion of its residues by boundedly many disjoint subsets $X_{j,l}$, each of positive parameter-dependent relative size and bounded doubling in $\mathbb{Z}/q\mathbb{Z}$. Lift each one over the entire vertical interval of the slab. The lifted pieces are disjoint, Lemma 7.4 bounds their integer doubling, and the omitted residue proportion equals the omitted point proportion. Since every retained slab has size at least $\varepsilon n/(4v)$, each lifted piece is linear in n . The additional omitted mass is at most $\varepsilon n/4$. This proves the stated conclusion, with room in the error budget. $\square$

Remark 7.6 (A stronger representation inside this terminal model). Green–Sanders [8] gives an additional description here: $\mathbf{1}_C$ is an exact signed sum of $O_{K_0}(1)$ arithmetic-progression indicators. Indeed each $\mathbf{1}_{X_j}$ has bounded finite-group algebra norm and is a bounded signed

sum of coset indicators. A coset in $\mathbb{Z}/q\mathbb{Z}$ has representatives $r + d\{0, \dots, q/d - 1\}$, $d \mid q$, $0 \leq r < d$. Its lift over a vertical interval $[a, a + L)$ is the single progression

$$r + qa + d\{0, \dots, (q/d)L - 1\}.$$

Lifting the signed identities slab by slab gives the assertion. This observation is not needed for Theorem 1.5. Nor does it bound the length of each progression by $|C|$: large progressions in a signed representation may cancel. Thus it should not be identified with the full signed-progression conjecture in [2].

8. A DEPTH-INDEPENDENT SIDON BUDGET AND THE INVERSE THEOREM

The local covering theorems are not sufficient by themselves for an arbitrarily deep assembly. A multiplicative loss at every split could vanish along a long path. We use a weaker coefficient exponent with constant exactly one, and then sum over all leaves at once.

8.1. A coefficient inequality with constant one.

Lemma 8.1 (Constant-one Sidon aggregation). *Let R be a finite additive Sidon set in the dual of a compact abelian group. For arbitrary complex coefficients,*

$$\left\| \sum_{r \in R} a_r \chi_r \right\|_1 \geq \left(\sum_{r \in R} |a_r|^{20} \right)^{1/20}. \quad (8.1)$$

The Haar measure is normalized. The exponent 20 is not asserted to be optimal.

Proof. The zero polynomial is immediate. Divide by a coefficient of maximum absolute value and multiply by the conjugate of its character. This preserves both sides up to the same factor, preserves the additive Sidon property, and leaves a polynomial $1 + g$ whose other coefficients have absolute value at most one. Let their squared absolute values sum to t . The case $t = 0$ is immediate, so assume $t > 0$.

Orthogonality and the Sidon condition give

$$\mathbb{E}g = \mathbb{E}g^2 = 0, \quad \mathbb{E}|g|^2 = t, \quad \mathbb{E}|g|^4 \leq 2t^2. \quad (8.2)$$

For example, $\mathbb{E}g^2 = 0$ follows because the sum of two nonzero shifted labels cannot equal the sum $0 + 0$ of the distinguished label. For the fourth moment only the two trivial pairings survive, with the all-equal terms counted only once. In particular, if $y = \operatorname{Im} g$, then $\mathbb{E}y^2 = t/2$.

For every $z \in \mathbb{C}$ one has

$$|1 + z| - 1 - \operatorname{Re} z \geq \frac{(\operatorname{Im} z)^2}{2(1 + |z|)}. \quad (8.3)$$

When $\operatorname{Im} z \neq 0$, rationalize the left side and bound its denominator $|1 + z| + 1 + \operatorname{Re} z$ above by $2(1 + |z|)$; when $\operatorname{Im} z = 0$, the right side vanishes and the inequality is automatic. Weighted Cauchy–Schwarz and (8.2) yield

$$\begin{aligned} \mathbb{E}|1 + g| &\geq 1 + \frac{(\mathbb{E}y^2)^2}{2\mathbb{E}[y^2(1 + |g|)]} \\ &\geq 1 + \frac{t}{4 + 8\sqrt{2t}}. \end{aligned} \quad (8.4)$$

For the second line, use $\mathbb{E}y^2|g| \leq \mathbb{E}|g|^3 \leq \sqrt{\mathbb{E}|g|^2\mathbb{E}|g|^4} \leq \sqrt{2}t^{3/2}$.

If $t \leq 2$, (8.4) is at least $1 + t/20 \geq (1 + t)^{1/20}$. If $t \geq 2$, apply L^1 – L^2 – L^4 interpolation to the whole Sidon polynomial:

$$\mathbb{E}|1 + g| \geq \frac{\|1 + g\|_2^3}{\|1 + g\|_4^2} \geq \sqrt{\frac{1 + t}{2}} \geq (1 + t)^{1/20}.$$

The last inequality holds for $1 + t \geq 3$. Finally, since each normalized coefficient has modulus at most one, its twentieth power is no larger than its square. Thus the normalized right side of (8.1) is at most $(1 + t)^{1/20}$, proving the lemma. $\square$

Proposition 8.2 (Fibre and leaf budgets). *At any internal vertex of Definition 1.4,*

$$\mathcal{L}(A_v)^{20} \geq \sum_{r \in R_v} \mathcal{L}(A_{v,r})^{20}. \quad (8.5)$$

Consequently, for an assembly tree with leaves A_l ,

$$\sum_l \mathcal{L}(A_l)^{20} \leq \mathcal{L}(A)^{20}. \quad (8.6)$$

If $|A| = N$ and $\mathcal{L}(A) \leq K \log N$, then for every fixed $0 < \theta < 1$,

$$\#\{l\} \leq (K \log N)^{20}, \quad (8.7)$$

$$\#\{l : |A_l| \geq N^\theta\} \leq \left(\frac{K}{c_{\text{MPS}} \theta} \right)^{20}, \quad (8.8)$$

$$\sum_{|A_l| < N^\theta} |A_l| \leq N^\theta (K \log N)^{20} = o_K(N). \quad (8.9)$$

Proof. Affine normalization preserves all relevant norms. For the normalized vertex, write $\xi = (j + x)/q_v$. At fixed x its Fourier polynomial is

$$F_{A_v}((j + x)/q_v) = \sum_{r \in R_v} e(rx/q_v) F_{A_{v,r}}(x) e(rj/q_v).$$

Apply Lemma 8.1 on the finite cyclic group in j , and then integrate over $x \in [0, 1]$. Minkowski's inequality in the finite space $\ell^{20}(R_v)$ gives

$$\begin{aligned} \mathcal{L}(A_v) &\geq \int_0^1 \left(\sum_r |F_{A_{v,r}}(x)|^{20} \right)^{1/20} dx \\ &\geq \left(\sum_r \mathcal{L}(A_{v,r})^{20} \right)^{1/20}. \end{aligned}$$

This proves (8.5). Induction on the finite tree proves (8.6), without a factor depending on its depth.

Every nonempty leaf has norm at least one, proving (8.7). The weighted MPS bound gives $\mathcal{L}(A_l) \geq c_{\text{MPS}} \log(|A_l| + 1) \geq c_{\text{MPS}} \theta \log N$ for a large leaf. Substitute this in (8.6) to obtain (8.8). Finally the number of small leaves times N^θ bounds their total cardinality, proving (8.9). Their affine images in A are disjoint, so these cardinalities are also their actual masses in the root set. $\square$

8.2. Proof of the almost-covering theorem.

Proof of Theorem 1.5 for $A \in \mathfrak{G}_{\beta, \Delta}$. Let

$$B_0 = \max \left\{ 1, \left(\frac{2K}{c_{\text{MPS}}} \right)^{20} \right\}.$$

Use Proposition 8.2 with $\theta = 1/2$. There are at most B_0 leaves of size at least $\sqrt{N}$. All other leaves together have size at most $\sqrt{N}(K \log N)^{20}$. Choose N_0 large enough that this quantity is at most $\varepsilon N/4$ for $N \geq N_0$, and discard these small leaves.

Among the remaining leaves discard any of size less than $\varepsilon N/(4B_0)$. Since there are at most B_0 of them, this removes at most another $\varepsilon N/4$ points. Every retained leaf has

$$n_l := |A_l| \geq \frac{\varepsilon N}{4B_0}, \quad \mathcal{L}(A_l) \leq \mathcal{L}(A) \leq K \log N \leq 2K \log n_l \quad (8.10)$$

for sufficiently large N , depending only on K, ε . Enlarge N_0 also so that these leaves exceed the size thresholds in Theorem 7.5. They are not singletons. Their affine normalizations are one of the two terminal geometries.

Apply Theorem 6.4 or Theorem 7.5 to each retained leaf, with norm parameter $2K$ and local error $\varepsilon/2$. The sum of these local exceptional masses is at most $\varepsilon N/2$. There are at most B_0 leaves and a bounded number of pieces per leaf. Each piece is a fixed parameter-dependent fraction of its leaf, and by (8.10) therefore a fixed fraction of N .

Lift the pieces through the affine maps along their paths to the root. These maps preserve cardinality and sumset doubling. Distinct lifted pieces remain disjoint, because the child images partition each parent. The total omitted mass is at most $\varepsilon N/4 + \varepsilon N/4 + \varepsilon N/2 = \varepsilon N$. This proves (1.12), with constants depending only on $K, \beta, \Delta, \varepsilon$. The Freiman-model extension will be proved in Section 9. $\square$

Proof of Corollary 1.6. Apply Theorem 1.5 with error $\varepsilon = \delta/4$. Denote its number of pieces by J and its doubling bound by Q . If $Y \subseteq A$ has $|Y| \geq \delta N$, then at least $\delta N/2$ of its points lie in the pieces. Thus, for some j , $X = Y \cap B_j$ has size at least $\delta N/(2J)$. Since $|X + X| \leq |B_j + B_j| \leq QN$, (2.2) gives

$$E(Y) \geq E(X) \geq \frac{|X|^4}{QN} \geq \frac{\delta^4}{16QJ^4} N^3 \geq \frac{\delta^4}{16QJ^4} |Y|^3.$$

The lower bound for the constant depends only on the stated parameters. $\square$

Remark 8.3 (Why the hereditary formulation is natural). For a uniform family of sets, the almost-covering conclusion for every fixed $\varepsilon > 0$ is qualitatively equivalent to positive normalized energy for every subset of fixed positive density. One direction is the preceding proof. For the converse, while a remainder has at least εN elements, use its energy lower bound and BSG to remove at least a fixed positive fraction of the original N . This terminates in boundedly many steps. No monotonicity of the Fourier L^1 norm under deletion is part of either argument.

9. HIGH-ORDER MODELLING AND SMALL EXCEPTIONAL PERTURBATIONS

An order-two Freiman isomorphism preserves additive energy and sumset cardinality, but does not generally preserve a Fourier L^1 norm. For the transport used here we retain sufficiently many moments. This also provides a bound for every corresponding subset simultaneously.

Definition 9.1 (Freiman isomorphism of order h). A bijection $\phi : A \rightarrow \tilde{A}$ between finite subsets of abelian groups is a Freiman isomorphism of order h if

$$a_1 + \cdots + a_h = b_1 + \cdots + b_h \iff \phi(a_1) + \cdots + \phi(a_h) = \phi(b_1) + \cdots + \phi(b_h)$$

for all choices with repetitions allowed. For nonempty sets, padding both sides with the same element shows that this condition also preserves equal-length sum relations of every order at most h .

Lemma 9.2 (Uniform transport of the norm). *Let $h \geq 2$, and let $\phi : A \rightarrow \tilde{A}$ be a Freiman isomorphism of order h . For every $C \subseteq A$, with $m = |C|$,*

$$|\mathcal{L}(C) - \mathcal{L}(\phi(C))| \leq \frac{4m}{\pi(2h+1)}. \quad (9.1)$$

Here the norms are taken on the corresponding compact dual groups. Moreover,

$$E(C) = E(\phi(C)), \quad |C + C| = |\phi(C) + \phi(C)|. \quad (9.2)$$

Proof. The empty-set case is immediate. The relation hypothesis and orthogonality give

$$\int |F_C|^{2j} = \int |F_{\phi(C)}|^{2j} \quad (1 \leq j \leq h).$$

For $-1 \leq t \leq 1$, the absolutely convergent Chebyshev expansion is

$$|t| = \frac{2}{\pi} + \frac{4}{\pi} \sum_{k \geq 1} \frac{(-1)^{k+1} T_{2k}(t)}{4k^2 - 1}. \quad (9.3)$$

One may obtain this by substituting $t = \cos \theta$ and computing the cosine coefficients of $|\cos \theta|$. The tail after $k = h$ is bounded uniformly by

$$\frac{4}{\pi} \sum_{k > h} \frac{1}{4k^2 - 1} = \frac{2}{\pi(2h + 1)},$$

using the telescoping identity $(4k^2 - 1)^{-1} = \frac{1}{2}((2k - 1)^{-1} - (2k + 1)^{-1})$. The truncation of (9.3) is an even polynomial, hence a polynomial of degree h in t^2 . Apply it to $|F_C|/m$ and to $|F_{\phi(C)}|/m$. The integrals of all polynomial terms agree, and the two approximation errors together are at most $4/[\pi(2h + 1)]$. Multiplying by m proves (9.1). The energy equality is the second moment identity above. For sumsets, map $a + b$ to $\phi(a) + \phi(b)$: the order-two equivalence makes this map well-defined and bijective, proving the other assertion. $\square$

Completion of Theorem 1.5 in a high-order model. Suppose $|A| = N \geq 2$ and $\phi : A \rightarrow \tilde{A}$ is an order- N Freiman isomorphism with $\tilde{A} \in \mathfrak{G}_{\beta, \Delta}$. Lemma 9.2 gives

$$\mathcal{L}(\tilde{A}) \leq \mathcal{L}(A) + 2/\pi \leq (K + 1) \log N.$$

Apply the already established theorem to $\tilde{A}$ with parameter $K + 1$. Pulling the pieces back by ϕ preserves their cardinalities, disjointness, covering error, and doubling constants by (9.2). This proves the stated extension. $\square$

Remark 9.3 (The budget is not charged once per model change). If successive models all belong to the same order- N relation class, Lemma 9.2 compares any two stages directly. The uniform error for every corresponding subset is less than $2/\pi$, regardless of the number of changes. This is the relevant difference from applying a rough norm-distortion estimate separately at each step. It does not assert that a geometric assembly can always be found in that relation class. Minimal-model arguments, such as those developed by Bedert [1], address the size of an integer representative; that is a separate issue from membership in $\mathfrak{G}_{\beta, \Delta}$.

Corollary 9.4 (Small exceptional perturbations). *Fix $H_0 \geq 0$. Suppose $|A| = N$, $\mathcal{L}(A) \leq K \log N$, and there is $A_0 \in \mathfrak{G}_{\beta, \Delta}$, or a set with a corresponding order- $|A_0|$ model in that class, such that*

$$|A \triangle A_0| \leq H_0 (\log N)^2.$$

Then the conclusion of Theorem 1.5 holds for A , with constants also allowed to depend on H_0 .

Proof. Put $s = |A \triangle A_0|$. The nonzero Fourier coefficients of $F_A - F_{A_0}$ are signs on the symmetric difference, so Parseval gives

$$|\mathcal{L}(A) - \mathcal{L}(A_0)| \leq \|F_A - F_{A_0}\|_1 \leq \sqrt{s}.$$

Thus $|A_0| = N + o(N)$ and $\mathcal{L}(A_0) \leq K' \log |A_0|$ for a fixed $K' = O(1 + K + \sqrt{H_0})$. Apply Theorem 1.5 to A_0 , using, for example, error $\varepsilon/4$. Intersect each resulting piece with A . At most $s = o(N)$ points are lost in total. Since the number of pieces is bounded and each originally has size at least cN , each intersection has at least half its former size for sufficiently large N . Its sumset cannot increase, so its doubling constant increases by at most a factor two. Points of $A \setminus A_0$, along with the original exceptional set, fit within the prescribed εN budget after increasing the size threshold. $\square$

10. EXAMPLES, PARAMETER OBSTRUCTIONS, AND THE GENERAL INVERSE PROBLEM

10.1. Interlaced progressions and the meaning of the energy exponent. The following example explains why the structural conclusion should not be formulated only through energetic initial segments. It is also a test that the assembly hypothesis alone is not already the conclusion.

Proposition 10.1 (Sidon fibres). *For all positive integers R, M , there is an integer set A of size $N = RM$ with*

$$\mathcal{L}(A) \leq \sqrt{R} \mathcal{L}(\{0, \dots, M-1\}), \quad (10.1)$$

and, for every $Y \subseteq A$,

$$E(Y) \leq 2M|Y|^2. \quad (10.2)$$

Every initial segment $P \subseteq A$ with $n = |P| \geq R$ also satisfies

$$E(P) \leq 16n^3/R. \quad (10.3)$$

The set belongs to $\mathfrak{S}_{\beta, \Delta}$ for every fixed $0 < \beta \leq 1$, $\Delta \geq 1$.

Proof. Take $B = \{1, 3, \dots, 3^{R-1}\}$, $q = 3^R$, and $A = B + q\{0, \dots, M-1\}$. The base-three expansion shows that B is Sidon as an integer set; since $q > 2 \max B$, it is also Sidon modulo q . In particular the displayed parametrization is disjoint. Writing $D_M(x) = \sum_{j=0}^{M-1} e(jx)$, finite Parseval gives, for fixed x ,

$$\frac{1}{q} \sum_{j=0}^{q-1} |F_B((j+x)/q)| \leq \left(\frac{1}{q} \sum_{j=0}^{q-1} |F_B((j+x)/q)|^2 \right)^{1/2} = \sqrt{R}.$$

Since $F_A(\theta) = F_B(\theta)D_M(q\theta)$, integration proves (10.1).

Write $Y = \bigcup_{b \in B} (b + qY_b)$, where $Y_b \subseteq \{0, \dots, M-1\}$. In any additive quadruple from Y , the residue labels on the two sides pair trivially. For two fibres,

$$E(Y_b, Y_c) := \int |F_{Y_b}|^2 |F_{Y_c}|^2 \leq \min\{|Y_b|, |Y_c|\} |Y_b| |Y_c| \leq M |Y_b| |Y_c|.$$

For the first inequality, use the supremum bound for one Fourier factor and Parseval for the other, and choose the smaller result. Summing over the two possible pairings of residue labels proves (10.2); the double count when all labels agree only makes an upper bound larger.

For an initial segment of size $n \geq R$, put $m = \lfloor n/R \rfloor$. All its points lie in $B + q\{0, \dots, m\}$. There are at most $2R^2$ ordered residue quadruples, and at most $(m+1)^3$ vertical solutions for each, because three vertical coordinates determine the fourth. As $m+1 \leq 2n/R$, this gives (10.3). Finally, the root has Sidon residue support B , and each child is an interval. Intervals are long-block terminals with $V = 1$, so A belongs to the stated class. $\square$

Taking $K = C\sqrt{R}$ in this example and letting M tend to infinity shows that no universal initial-segment statement can replace the K^{-2} scale by K^{-p} for a fixed $p < 2$, with a positive coefficient independent of K . For fixed $\delta > 0$, choose M sufficiently large that every initial segment of size $\gg N^{1-\delta}$ has size at least R , and then let $R \rightarrow \infty$ in (10.3). This is a statement about the parameter dependence of that statistic, not about extremizers for the sharp Littlewood constant.

Indeed each individual fibre is an arithmetic progression of length $M = N/R$, with energy $(2M^3 + M)/3$. For fixed K , selecting such a fibre already gives a linear-sized structured piece in this example. Conversely, (10.2) shows that as $R \rightarrow \infty$ the class membership alone cannot give uniform positive-density small-doubling pieces: if $|Y| \geq cN$ and $|Y + Y| \leq Q|Y|$, then (2.2) forces $Q \geq cR/2$. The low- L^1 assumption is therefore essential to Theorem 1.5.

There is also a parameter restriction for statements that allow arbitrary subsets. The same example gives $|Y|\omega(Y) \leq 2N/R$. If both K and R grow with N , any proposed improvement must account for the corresponding size cost. This does not preclude a fixed- K linear-sized inverse theorem, which is the structural direction considered here.

10.2. Depth is genuinely unbounded. Let $h, M \geq 1$ and consider

$$A = 3^h \{0, \dots, M-1\} \cup \{3^j : 0 \leq j < h\}. \quad (10.4)$$

Successively reducing modulo 3 gives a chain of h Sidon splits with residue support $\{0, 1\} \subset \mathbb{Z}/3\mathbb{Z}$. At each step there is one continuing child and one singleton. The terminal continuing set is an interval. On the other hand, the triangle inequality and Parseval give

$$\mathcal{L}(A) \leq \mathcal{L}(\{0, \dots, M-1\}) + \sqrt{h}.$$

For $h = \lfloor (\log M)^2 \rfloor$, this is $O(\log |A|)$ while the depth tends to infinity. A per-step multiplicative loss would not suffice for this family. Proposition 8.2 has no such loss.

For comparison, the fully branching ternary digit set

$$\left\{ \sum_{j=0}^{h-1} \epsilon_j 3^j : \epsilon_j \in \{0, 1\} \right\}$$

has 2^h singleton leaves in a Sidon tree. The leaf budget implies $\mathcal{L}(A) \geq |A|^{1/20}$, incompatible with $K \log |A|$ for large $|A|$ and fixed K . Thus the same estimate distinguishes many macroscopic independent branches from a long chain with only small exceptional branches.

10.3. What the proof does not supply. The missing step for an unrestricted inverse theorem is not part of BSG, the majority calculation, or the leaf budget. It is a theorem placing arbitrary low- L^1 sets, or suitable high-order equivalent models, into a usable assembly. Non-Sidon quotient supports need not be dense separated blocks or long arithmetic inflations. The absence of a Sidon split is not a classification of a general integer set.

A related difficulty is the difference between localizing a test function and localizing the target. The kernel V_D in Lemma 6.2 has $2D+1$ Fourier coefficients equal to one but $\|V_D\|_1 \leq 3$, because its other coefficients form fractional slopes. Thus retaining many unit coefficients in a smooth projection does not justify replacing that projection by an integer indicator with comparable norm. All target sets in the local theorems above are actual integer sets, and all residue supports at internal vertices are the full supports. These requirements are used explicitly.

A natural unrestricted target would assert (1.12) for every A with $\mathcal{L}(A) \leq K \log N$. It would imply a linear-sized small-doubling subset and positive normalized energy for every macroscopic subset. The signed-progression conjecture formulated in [2, Section 6] asks for a still more specific representation, with boundedly many progressions and a sublinear exceptional set. Neither general statement is proved here. The present theorem isolates local mechanisms that are compatible with arbitrarily deep independent assemblies, and makes their exact entry hypotheses explicit.

APPENDIX A. AN EXACT CERTIFICATE FOR THE NUMERICAL COEFFICIENT

This appendix specifies a finite rational computation certifying the decimal in Theorem 1.1. Decimal evaluations are used only for presentation. Every inequality required by the certificate is an inequality between rational numbers, or follows from squaring two nonnegative rational bounds.

The even moments I_{2r} are given exactly by (3.4). For the odd coefficients, choose an upper rational square-root bound as follows. If $x \geq 0$ is rational, let

$$U_d(x) = \frac{\lfloor 10^d \sqrt{x} \rfloor + 1}{10^d}.$$

The integer in this definition is computed by integer square root after clearing denominators. Then $\sqrt{x} < U_d(x)$, with an error at most 10^{-d} . Set $\bar{c}_{2r} = I_{2r}$ and $\bar{c}_{2r+1} = U_{40}(I_{2r}I_{2r+2})$. Using $c_k \leq 1$, for any $L \geq 5$ one has

$$\sum_{k \geq 5} \frac{c_k a^k}{k} \leq \sum_{k=5}^L \frac{\bar{c}_k a^k}{k} + \frac{a^{L+1}}{(L+1)(1-a)}. \quad (\text{A.1})$$

We use $L = 160$ and $a = 9/10$.

Logarithms are bounded by their elementary positive series. For a rational $x > 1$, write $z = (x-1)/(x+1)$ and

$$\ell_J(x) = 2 \sum_{j=0}^{J-1} \frac{z^{2j+1}}{2j+1}.$$

Then

$$\ell_J(x) < \log x < \ell_J(x) + \frac{2z^{2J+1}}{(2J+1)(1-z^2)}. \quad (\text{A.2})$$

The certificate takes $J = 240$ for $x = 9, 10$. These estimates follow by integrating the geometric series for $(1-z^2)^{-1}$ and bounding each tail denominator below by $2J+1$.

Substituting (A.1) into (1.5), with all square roots rounded upwards, gives a rational number $\bar{\mathcal{E}}$ satisfying

$$\mathcal{E}_{9/10} \leq \bar{\mathcal{E}} < \frac{479084457020255861}{10^{18}}. \quad (\text{A.3})$$

The series remainder in (A.1) is less than $2.668198549256 \cdot 10^{-9}$. Its contribution to $\bar{\mathcal{E}}$ is multiplied by $2/9$. Let $\bar{L}_9$ be the upper bound for $\log 9$ in (A.2), and let $\bar{R} = U_{40}(\bar{\mathcal{E}}/3)$. Then

$$c_* \geq \frac{9}{10\bar{L}_9}(1 - \bar{R}) > \frac{2459209213577253}{10^{16}} > 0.2459209. \quad (\text{A.4})$$

The last inequality is checked by integer cross-multiplication; its numerator $1 - \bar{R}$ is positive. The same logarithmic enclosure verifies that (4.16) at $b = 11/8$ is negative, with an upper bound less than -1.1121923140 .

The companion file `verify_constant.py` implements these steps using integer arithmetic, rational fractions, and integer square roots. Its printed decimal strings are not inputs to the certificate. The source and the certified rational procedure are supplied to make the explicit constant reproducible, rather than to replace the analytical argument preceding this appendix.

APPENDIX B. THE ONE-SIDED ENDPOINT BARRIER AND A TWO-SIDED INTERVAL MODEL

This appendix distinguishes the sharp Littlewood constant from the best value obtainable in the final spectral class of the outer recursion. The upper bound below is the classical Landau inequality; see [13, 12]. We include its short proof to make the limitation of the method precise.

Proposition B.1 (Landau's bound in the relevant spectral class). *Let $D_N(z) = 1 + z + \dots + z^{N-1}$ on the unit circle. If $\|G\|_\infty \leq 1$ and $\text{supp } \widehat{G} \subseteq (-\infty, N-1] \cap \mathbb{Z}$, then*

$$|\langle D_N, G \rangle| \leq \sum_{k=0}^{N-1} \left(\frac{\binom{2k}{k}}{4^k} \right)^2 = \frac{1}{\pi} \log N + O(1). \quad (\text{B.1})$$

The inequality is sharp within this class of tests.

Proof. Put $m = N-1$, $b_k = \binom{2k}{k}/4^k$, and $Q_m(z) = \sum_{k=0}^m b_k z^k$. Since the b_k are the coefficients of $(1-z)^{-1/2}$,

$$Q_m(z)^2 = 1 + z + \dots + z^m + O(z^{m+1}).$$

The boundary function $h = z^m \overline{Q_m}$ is analytic and bounded by one in the disc. If h_k are its Taylor coefficients, then

$$\langle D_N, G \rangle = \sum_{k=0}^m h_k = [z^m](h Q_m^2).$$

Taking this coefficient on the unit circle gives

$$|[z^m](h Q_m^2)| \leq \int_{\mathbb{T}} |Q_m|^2 = \sum_{k=0}^m b_k^2.$$

Stirling's formula gives $b_k^2 = 1/(\pi k) + O(k^{-2})$ for $k \geq 1$, proving the asymptotic.

For sharpness, Q_m has no zeros in the closed disc. This is immediate for $m = 0$. For $m \geq 1$, the identity

$$(1-z)Q_m(z) = 1 - \sum_{k=1}^m (b_{k-1} - b_k)z^k - b_m z^{m+1}$$

has positive subtracted coefficients summing to one. Their sum has modulus strictly less than one when $|z| < 1$. Equality on the unit circle can produce a zero of the right side only at $z = 1$, since the coefficient of z is positive; but $Q_m(1) > 0$. Thus $h = z^m \overline{Q_m}/Q_m$, with the numerator interpreted as the reversed polynomial, is analytic and has unit boundary modulus. It attains the bound, since $h Q_m^2 = z^m |Q_m|^2$ on the circle. The corresponding test is $G = Q_m/\overline{Q_m}$. $\square$

For an interval, every test produced in (4.4) has Fourier support at most $N-1$. Consequently modifying source phases, scale choices, or multiplier estimates while retaining that final support restriction cannot prove a coefficient greater than $1/\pi$. This is not an upper bound for the actual Littlewood constant, and it does not prohibit using one-sided tests as inputs to a nonlinear construction with two-sided spectrum.

Proposition B.2 (Two-sided completion on an interval). *With Q_{N-1} as above, put*

$$U_N = \frac{Q_{N-1}}{Q_{N-1}}, \quad V_N = z^{N-1} \overline{U_N}, \quad H_N = \frac{U_N + V_N}{|U_N + V_N|},$$

where $H_N = 0$ at zeros of its denominator. Then $|H_N| \leq 1$ and

$$\text{Re} \langle D_N, H_N \rangle \geq \|D_N\|_1 - O(1) = \frac{4}{\pi^2} \log N + O(1). \quad (\text{B.2})$$

Proof. In this proof write $z = e^{it}$ with $0 < t < 2\pi$; circle integrals carry measure $dt/(2\pi)$. Define

$$U_\infty(t) = ie^{-it/2}, \quad V_\infty(t) = -ie^{i(N-1/2)t}.$$

The identity

$$U_\infty(t) + V_\infty(t) = 2\sin(t/2)D_N(e^{it}) \quad (\text{B.3})$$

shows that the phase of this sum equals the phase of D_N . These are boundary phase models, not assertions about their one-sided Fourier support.

Abel summation applied to the decreasing coefficients b_k gives, on the unit circle away from 1,

$$|Q_{N-1}(z) - (1-z)^{-1/2}| \leq CN^{-1/2}|1-z|^{-1}.$$

For example the tail of $\sum_{k \geq N} b_k z^k$ is bounded by a constant times $b_N/|1-z|$ by summation by parts. Since $|(1-z)^{-1/2}| = |1-z|^{-1/2}$, normalization of complex numbers implies, with $s = \min(t, 2\pi - t)$,

$$|U_N - U_\infty| + |V_N - V_\infty| \leq C \min\{1, (Ns)^{-1/2}\}. \quad (\text{B.4})$$

Here the phase of $(1 - e^{it})^{-1/2}$ divided by its conjugate is exactly $ie^{-it/2}$.

On the arcs $s \leq 1/N$, use $|D_N| \leq N$ and the trivial bound two for the difference of normalized phases. The weighted error integral is $O(1)$. Outside these arcs, use

$$\left| \frac{X}{|X|} - \frac{Y}{|Y|} \right| \leq \frac{2|X - Y|}{|Y|}, \quad \frac{|D_N|}{|U_\infty + V_\infty|} = \frac{1}{2\sin(t/2)}.$$

At zeros the inequality is interpreted by an arbitrary unit-bounded value, and the exceptional finite set has measure zero. Combining with (B.4), the remaining weighted error is at most

$$CN^{-1/2} \int_{1/N}^{\pi} s^{-3/2} ds = O(1).$$

This proves the first inequality in (B.2); the second is (1.2). $\square$

For a general frequency set, nonlinear phase completion introduces mixed products such as $U^2\bar{V}$ and $V^2\bar{U}$ in the pairing. Opposite one-sided Fourier supports alone do not eliminate these terms or control their signs. Proposition B.2 is an interval model, not a proof of the sharp coefficient for arbitrary sets.

COMPUTATIONAL REPRODUCIBILITY AND ASSISTANCE

The explicit decimal is certified by the rational procedure in Appendix A; accompanying finite checks of the moment and Sidon identities are supplementary tests, not substitutes for the proofs. Generative-AI tools were used substantially in developing and testing arguments, organizing the manuscript, and preparing the exposition and code. Such assistance does not constitute an independent verification of the mathematics. Responsibility for the submitted manuscript rests with the author.

REFERENCES

- [1] B. Bedert, *Large sum-free subsets of sets of integers via L^1 -estimates for trigonometric series*, preprint (2025), [arXiv:2502.08624](#).
- [2] T. F. Bloom and B. Green, *Remarks on the inverse Littlewood conjecture*, Q. J. Math. (2026), article haag016, [doi:10.1093/qmath/haag016](#); [arXiv:2602.16482v2](#) (revised April 18, 2026).
- [3] J. Bourgain, *Estimates related to sumfree subsets of sets of integers*, Israel J. Math. **97** (1997), 71–92.
- [4] N. G. de Bruijn, C. van Ebbenhorst Tengbergen, and D. Kruyswijk, *On the set of divisors of a number*, Nieuw Arch. Wisk. (2) **23** (1951), 191–193.
- [5] P. L. Duren, *Theory of H^p Spaces*, Pure and Applied Mathematics, vol. 38, Academic Press, New York–London, 1970.
- [6] J. J. F. Fournier, *Some remarks on the recent proofs of the Littlewood conjecture*, in *Second Edmonton Conference on Approximation Theory* (Edmonton, Alta., 1982), CMS Conf. Proc., vol. 3, American Mathematical Society, Providence, RI, 1983, 157–170.
- [7] R. M. Gabriel, *The rearrangement of positive Fourier coefficients*, Proc. London Math. Soc. (2) **33** (1932), 32–51, [doi:10.1112/plms/s2-33.1.32](#).
- [8] B. Green and T. Sanders, *A quantitative version of the idempotent theorem in harmonic analysis*, Ann. of Math. (2) **168** (2008), no. 3, 1025–1054, [doi:10.4007/annals.2008.168.1025](#).
- [9] B. Hanson, *Littlewood’s problem for sets with multidimensional structure*, Int. Math. Res. Not. IMRN **2021** (2021), no. 21, 16736–16750, [doi:10.1093/imrn/rnaa043](#); [arXiv:2003.01561](#).
- [10] G. H. Hardy and J. E. Littlewood, *A new proof of a theorem on rearrangements*, J. London Math. Soc. **23** (1948), 163–168.
- [11] S. V. Konyagin, *On the Littlewood problem*, Izv. Akad. Nauk SSSR Ser. Mat. **45** (1981), no. 2, 243–265, 463; English translation, Math. USSR-Izv. **18** (1982), no. 2, 205–225.
- [12] L. V. Kovalev, *Sharp bounds for some segments of bounded power series*, preprint (2025), [arXiv:2507.04544](#).
- [13] Y. Li, S. Liu, S. Xu, and Y. Zhao, *Asymptotics of Landau constants with optimal error bounds*, Constr. Approx. **40** (2014), 281–305, [doi:10.1007/s00365-014-9259-x](#).
- [14] O. C. McGehee, L. Pigno, and B. Smith, *Hardy’s inequality and the L^1 norm of exponential sums*, Ann. of Math. (2) **113** (1981), no. 3, 613–618, [doi:10.2307/2007000](#).
- [15] G. Petridis, *The L^1 -norm of exponential sums in $\mathbb{Z}^d$* , Math. Proc. Cambridge Philos. Soc. **154** (2013), no. 3, 381–392, [doi:10.1017/S0305004112000588](#).
- [16] S. K. Pichorides, *On the L^1 norm of exponential sums*, Ann. Inst. Fourier (Grenoble) **30** (1980), no. 2, 79–89, [doi:10.5802/aif.785](#).
- [17] C. Reiher and T. Schoen, *Note on the theorem of Balog, Szemerédi, and Gowers*, Combinatorica **44** (2024), no. 3, 691–698, [doi:10.1007/s00493-024-00092-5](#); [arXiv:2308.10245v2](#).
- [18] J. D. Stegeman, *On the constant in the Littlewood problem*, Math. Ann. **261** (1982), no. 1, 51–54, [doi:10.1007/BF01456409](#).
- [19] T. Tao and V. H. Vu, *Additive Combinatorics*, Cambridge Studies in Advanced Mathematics, vol. 105, Cambridge University Press, Cambridge, 2006, [doi:10.1017/CB09780511755149](#).
- [20] K. Yabuta, *A remark on the Littlewood conjecture*, Bull. Fac. Sci. Ibaraki Univ. Ser. A **14** (1982), 19–21, [doi:10.5036/bfsiu1968.14.19](#).
- [21] A. Zygmund, *Trigonometric Series*, Vols. I and II, third edition, Cambridge Mathematical Library, Cambridge University Press, Cambridge, 2002.

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